Finite Element Modelling and Analysis of an Airplane Wing

MAE4426 Report 1

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Introduction

The purpose of this report is to document the modelling and analysis of a simplified aircraft wing. Finite element analysis will be conducted on an aircraft wing within the software ANSYS [1] to obtain stress and displacement data of the wing under operational loading.

Wing Choice

For this report a simplified elliptical wing was chosen, modelled after the wing of a Supermarine Spitfire. The Spitfire is a British single seater fighter aircraft that was mainly employed in the Royal Air Force throughout World War II. Its revolutionary low cross-section design made it the highest performing short-range interceptor of the time [2]. Another reason this wing was chosen is to simplify wing loading calculations, as an ideal elliptical loading can be applied. As seen in *figure 1* the internals of a spitfire wing are quite complex. To capture the behaviour of an elliptical spar-ribbed wing under loading, the internal structure will be simplified to save on modelling and computation time, while still capturing the required behaviour.



figure 1: Spitfire cutout

Relevant Geometry

The relevant dimensions of the forward-swept elliptical wing of a Spitfire can be summarised in figure 2. The internal geometry was derived from this wing planform area.

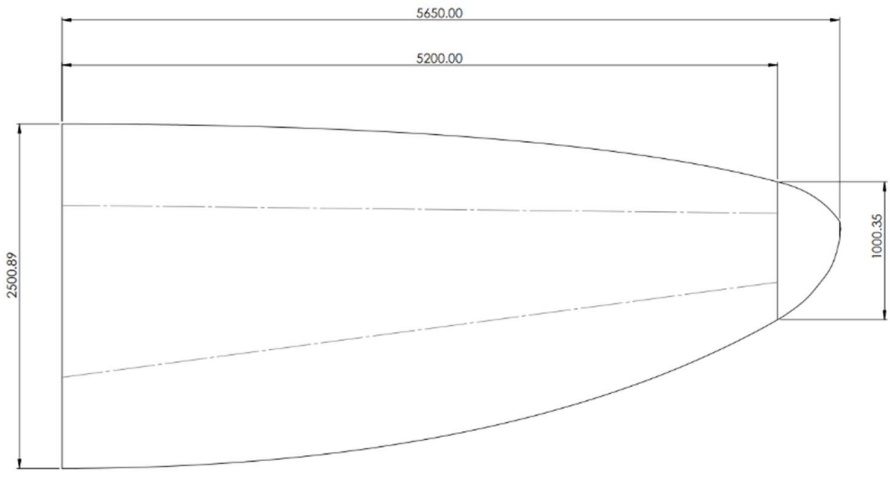


figure 2: Wing geometry

Relevant Specifications

For this simulation, all that is needed is the gross weight, an estimate of drag and a representative load case. The gross weight of a Spitfire is 3,004 kg [2] and a limit load manoeuvre would be approximately 6 g's of force [3].

Simplifications & Assumptions of Wing Geometry

To stay within a reasonable scope certain assumptions were made to reduce the complexity of the wing geometry and are summarised below.

- The **air foil profile** along the wing was collapsed into a single air foil. Originally a NACA 2213 at the root and a NACA 2209.4 at the tip, the profile was changed to include only the NACA 2213 along the entire profile. This reduces modelling complexity and the chance for erroneous FEA results along the change in profile. This would not affect the results significantly as the tip of the wing would see the least loading as expected with an elliptical loading profile. The change in profile near the tip was likely to counteract minor unfavourable aerodynamic effects caused by wing twisting [4].
- The **spars and rib** structures are simplified into one solid, with reduced geometry. As seen in figure 1 the internal structure of a spitfire wing is complex. The modelling of this structure is outside the scope of this project and representative behaviour can be obtained with simplified geometry. The simplified geometry will still contain rib cutouts to investigate mid-rib stresses on these features. The lack of joinery between the ribs and spars also removes the complexity of bolts and other attachment methods, which are outside the scope of this report.
- The **skin** of the wing will be omitted. The aim of this report is to study the behaviour of a wing under operational loading, which is predominantly bending stresses through the ribs and spars [5]. The skin on airplane wings plays the role of transferring aerodynamic loading to the underlying structure as well as playing a signficant role in torsional strength, the lack of this torsional strength will be noted in analysis.
- The **loading direction** will be considered normal to the planform of the wing. This would slightly differ to the case of loading through a slight angle due to the Spitfire having dihedral, which is the upwards angle of wings to provide lateral stability.
- The **spars** are modelled as approximately sized I-beams due to the lack of literature on the specifics of spitfire internal geometry.
- **Control surface** cutouts were removed from the design. As the main and rear spar of the Spitfire specifically are uninterrupted by the control cutouts, this will marginally affect the bending strength of the wing. The ribs that protrude into the region of control surfaces can be assumed to provide the same lift force that the control surfaces would in steady level flight.
- **Symmetrical** effects will be ignored, such as the change in stresses in the spar given the spars pass underneath the fuselage into both wings.

With all these assumptions stated, the wing structure can be modelled.

Modelling

The internal wing structure was modelled within SolidWorks. The assumptions were followed, and the final wing structure was designed based on the Spitfire's wing dimensions and reference images, this can be seen in figure 3.



figure 3: Wing structure render

The relevant spar and ribs thicknesses can be seen in figure 4.

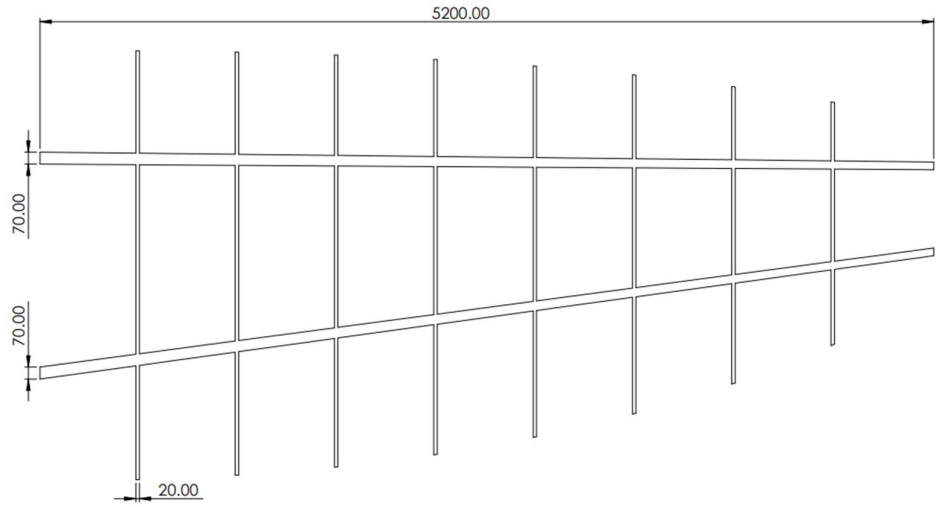


figure 4: Wing structure overview

Finite Element Modelling Process

Boundary and Loading Conditions

To model the performance of this wing, both spars will be considered as cantilever beams. Their root faces will be constrained in all 6 axes to represent the underbody attachment of the Spitfire.

The lift force will be applied to the lower outwards facing surfaces directly upwards. To ensure a representative loading, a pressure force will be applied to the bottom side of the ribs and spars in an appropriate magnitude to sum to the operational wing loadings. To calculate

the lift magnitude, the gross weight of 3,004 kg will be scaled by a maximum g manoeuvre considered for fighter pilots of 6 g's.

$$F_{wing} = \frac{1}{2} W_{wing}(6g) = \frac{1}{2}(3{,}004)(6)(9.81) = \ 88407.7\ N$$

This force will be distributed evenly along the bottom surface of the structure. It should be noted that a more accurate loading would be an elliptical distribution, although without a skin it becomes quite cumbersome to calculate an appropriate varying pressure field, and was considered outside the scope of this report. Using a simple force over area formulation, the following pressure force is obtained.

$$P_{underside} = \frac{F_{wing}}{A_{underside}} = \frac{88407.7}{0.93} = 95.062\ kN/m^2$$

The drag force will be applied along the upper and lower surfaces of the spars and ribs. This will be applied along the vector spanning from the leading to trailing edge. The magnitude of this force will be calculated for a steady-level flight case at the Spitfires maximum rated speed of 600 km/h. The drag equation given in *equation 1* relates the drag force to the air density (ρ), airspeed (V), coefficient of drag (C_D), and wing planform (S).

$$D \ = \ \frac{1}{2}\ \rho V^2 C_D S \quad (1)$$

The only unknown variable is the coefficient of drag. This can be calculated by using a prediction for the parasitic drag and lift induced drag. This was accomplished using a combination of simulation software and hand calculations that are detailed in *appendix A1*. The result was a drag coefficient of 0.0712, which led to a drag force of 3.392 kN, also detailed in the same appendix.

This load will also be represented via a pressure load; to calculate this the force will be distributed on all skin-interfacing surfaces.

$$P_{drag} = \frac{F_{drag}}{A_{skin}} = \frac{3392.31}{1.84} = 1843.65\ N/m^2$$

The loading and constraints can be seen summarised in Table 1 and figure 5.

Table 1: Loads and supports summary

Load/Support	Value	Location
Lift	95 kPa	Applied to underside skin contact faces
Drag	1.84 kPa	Applied to skin contact faces
Spar fixed support	N/A	Root face of spars

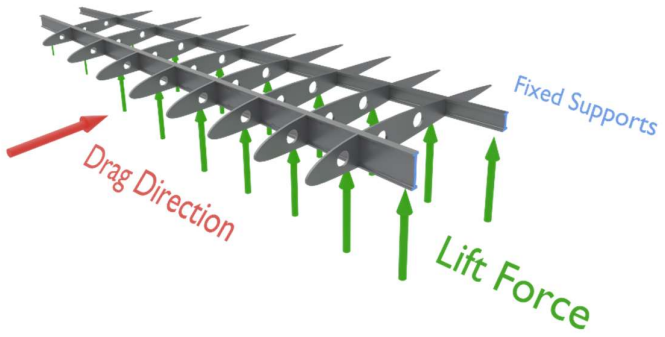


figure 5: Load case diagram

Material Choice

To best represent the metals present in the construction of the Supermarine Spitfire, Aluminium 2024-T4 alloy [6] was used. Also known as Duralumin, it is a high strength-to-weight ratio metal with excellent fatigue resistance. Composed largely of aluminium, copper, and magnesium, the alloy is an isotropic material and can be modelled within ANSYS using the values summarised in table 2.

table 2: Aluminium 2024-T4 properties

Property	Value
Modulus of Elasticity	73.1 GPa
Poisons Ratio	0.33
Tensile Yield Strength	324 MPa
Shear Modulus	28 GPa

Meshing Approach

To mesh the structure, the default ANSYS meshing values were used to begin with bar changing the element size to 30 mm. To attempt to capture the finer effects near the cutout sections of the wing, a local element sizing of 4 mm was applied. The initial mesh had 112,912 elements and can be seen in figure 6.

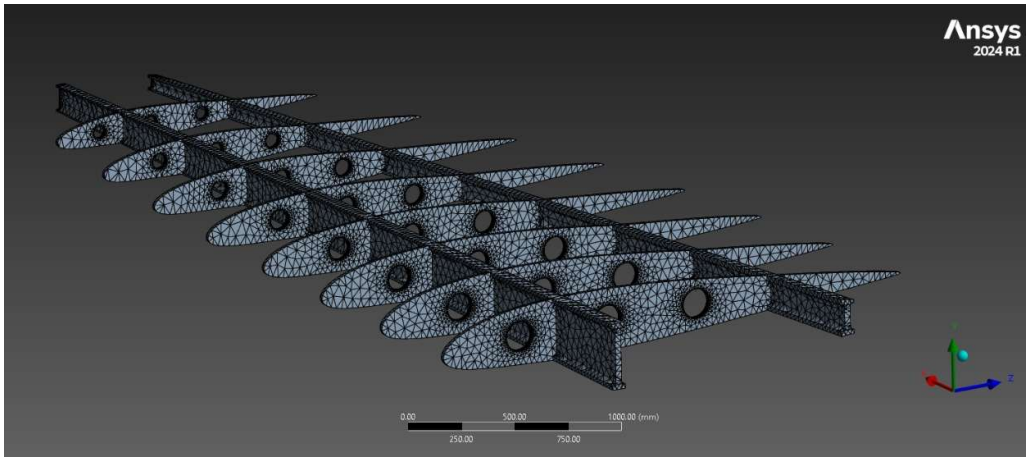


figure 6: Initial mesh

With this mesh a simulation to find the Von-Mises stress was ran. It resulted in an initial value of 317.25 MPa within the structure.

To verify this result, a mesh convergence study will be run. The principle is to increase the fineness of the mesh and monitor the resulting maximum stress at each level of mesh complexity. This is to ensure there are no stress concentrations that are being missed due to the lack of sufficient elements in that area. If the maximum stress plateaus with increase in element count, then it can be inferred the mesh is converged and sufficiently complex.

The results of increasing the mesh and solving for maximum Von-Mises's stress can be seen in figure 7.

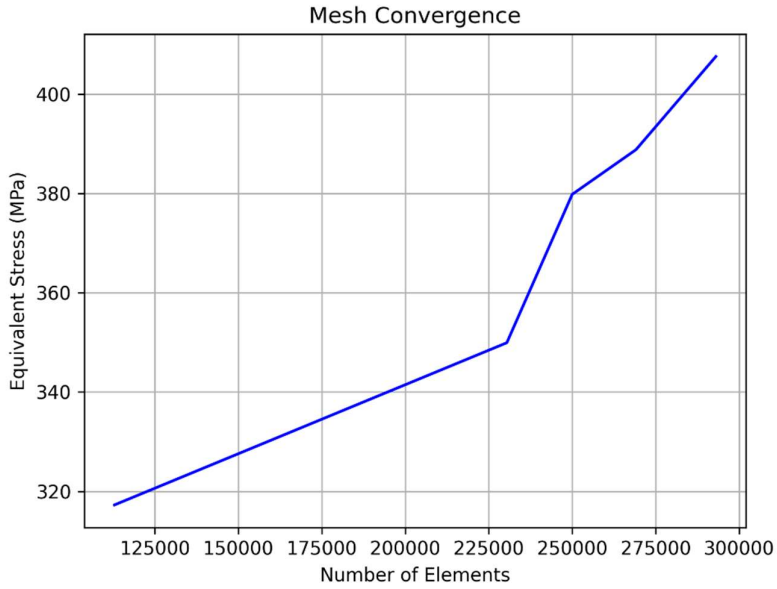


figure 7: Initial stress mesh convergence

It is evident the stress is not showing signs of converging. A closer analysis reveals a maximum stress concentration on an edge near the trailing edge spar as seen in figure 8.

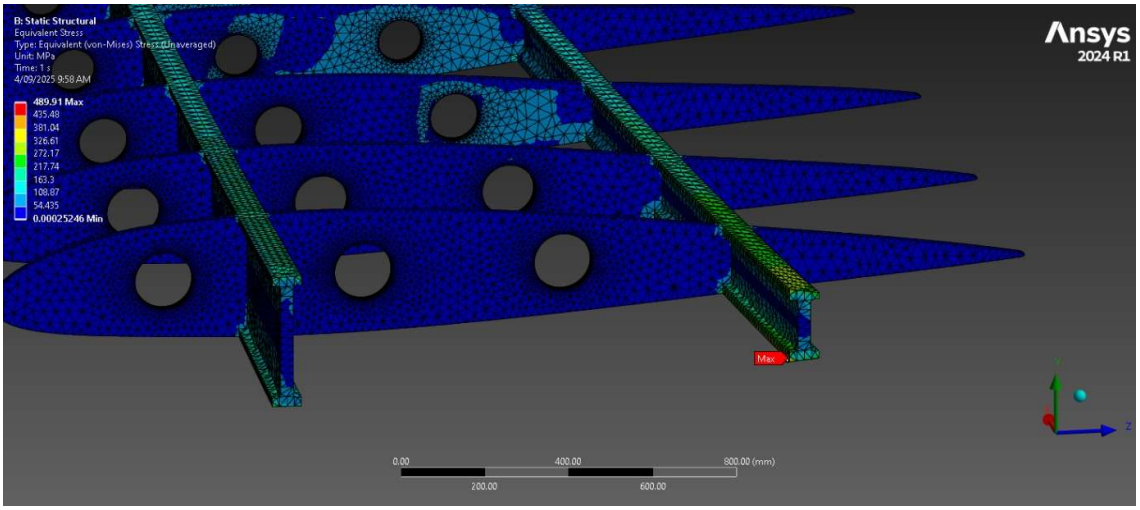


figure 8: Preliminary mesh stress solution

To combat this concentration in an area of complexity, the mesh will be simplified. Replacing the I-beams with slightly thicker rectangular spars will result in a more easily meshed shape. This will also allow quad elements to be used to better fit the structure. To aid in this transition the spars and ribs will no longer be considered one body, they will be split in two to create zones for meshing methods. The updated geometry and mesh can be seen in figure 9.

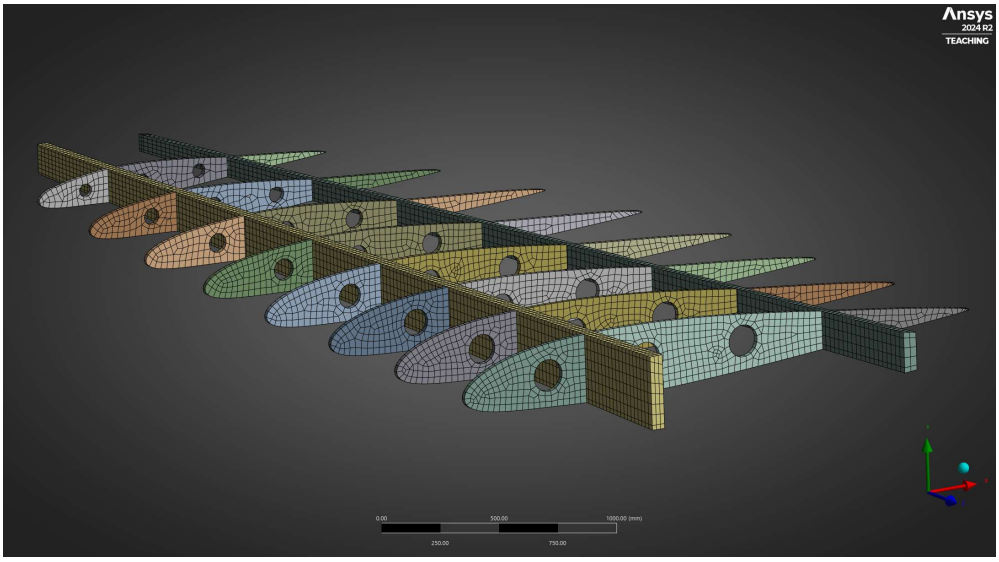


figure 9: Improved mesh with separate bodies

Quad elements flow along the span of the spars quite nicely and conform well around the rib cutouts and curvature. Qualitatively this should create a mesh that would avoid any stress concentrations around ill fit elements.

Another mesh convergence will be run on this geometry. This time average deformation will be used to converge the mesh, this will additionally avoid the inclusion of problematic concentrations. This convergence can be seen in figure 10.

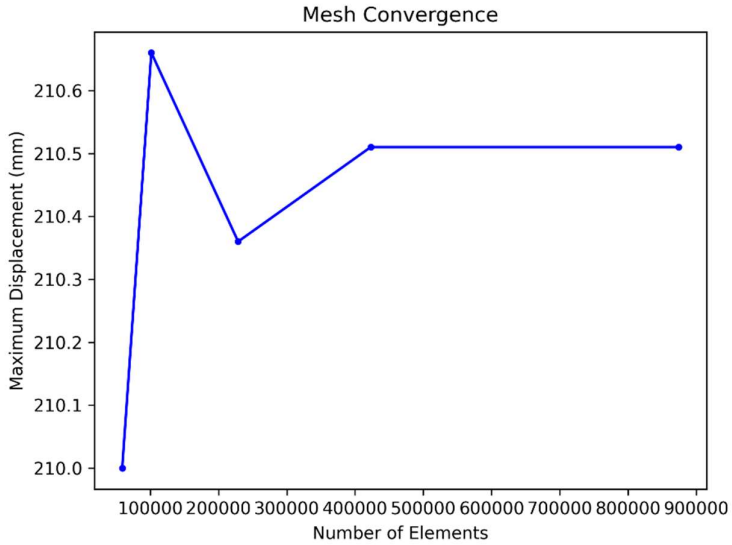


figure 10: Deformation mesh convergence

With this result it can be seen the simulated deformation has plateaued. This indicates convergence of the mesh as the results changed less than 1% between iterations. The finest mesh will be used to analyse the remaining results of the model.

Results

Stress Analysis

After the mesh was converged using the maximum deformation the equivalent Von-Mises stress was solved for and is displayed in figure 11, as well as the results summarised in table 4.

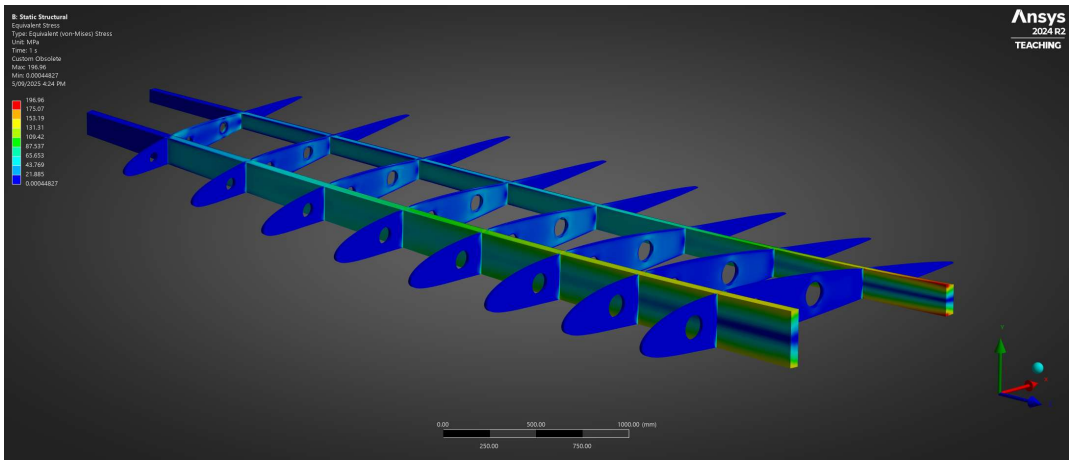


figure 11: Equivalent stress solution

table 3: Stress simulation results

Equivalent Stress (Von-Mises)	**Value**
Maximum	196 MPa
Minimum	448 Pa

Within a wing structure it is expected that the spars will carry most of the load throughout the wing in the form of a resisting bending moment at the root [5]. To better interpret the results, observe *equation 2* that gives the stress caused by a moment around a cross section with an area moment of inertia I.

$$\sigma = \frac{My}{I} \quad (2)$$

With y defined as the midplane of the beam, the expected loading will be positive upwards (tensile stress) and negative downwards (compressive stress) in a linear behaviour beginning at zero stress in the midplane [7].

This behaviour is evident in the stress distribution shown in figure 11, where the upper and lower surfaces of the spars experience the highest loads. These regions correspond to the critical failure locations, with maximum stress occurring at the lower surface of the beam due

to tensile loading. The concentration of stresses at these points arises from the boundary condition: the wing is constrained at the root, causing the aerodynamic lift forces to be reacted through bending moments that maintain structural stiffness under load.

In both figures, the rear spar is predicted to fail first, as the highest stress concentrations occur at its root. However, this outcome does not accurately reflect real aerodynamic loading. The applied load case simplifies the complex aerodynamic pressure distribution into a uniform pressure field, which artificially shifts a greater proportion of the load onto the trailing section of the wing. In reality, the lift generated by an air foil acts predominantly near the quarter-chord, resulting in a higher load transfer through the front spar than the rear [8]. This discrepancy should be noted when interpreting the results of the present simulation. Despite this, rear-biased force does show the role of the ribs in connecting both spars, as seen in figure 11 the ribs are resisting the torsional moment caused by the spars moving non uniformly which is critical to maintaining wing shape.

The root bending moment represents a major structural constraint in fixed-wing aircraft. As wings become longer or are subjected to higher load cases, the resulting increase in root moment necessitates additional reinforcement at the wing root. To mitigate this effect, the bending load must be more effectively distributed. Potential strategies include increasing the number of spars to enlarge the effective cross-sectional area and thereby reduce stress, shortening the wingspan to shift lift forces closer to the root, or incorporating auxiliary structures such as struts to provide additional support nearer to the wing tip.

Displacement Analysis

The displacement results can be seen in Figure 12 and Figure 13 as well as summarised in table 4.

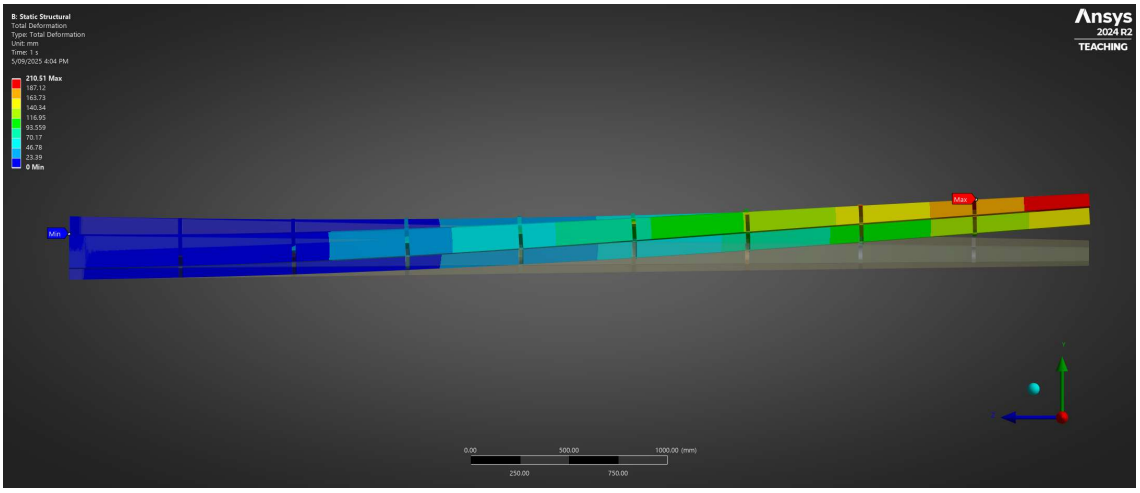


Figure 12: Displacement simulation

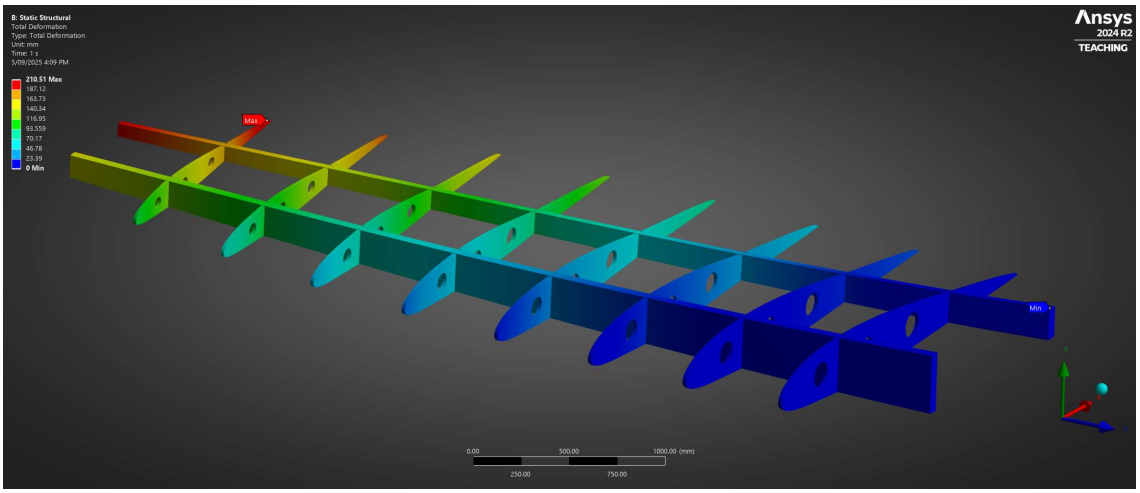


Figure 13: Displacement simulation

table 4: Deformation simulation results

Deformation	Value
Maximum	210.51 mm
Minimum	0 mm

The deformation is observed to increase with distance from the root, which is consistent with the expected behaviour of a wing under load. With the root constrained, deflection grows progressively along the span, reaching a maximum at the wing tip. In the simulation, the maximum displacement occurs at the trailing edge of the most outboard rib, resulting from a combination of tip deflection and torsional twisting of the wing. While some degree of twisting near the tips is expected due to the nose-down pitching moment generated by cambered air foils, the magnitude observed here is excessive and most likely attributable to the simplified loading assumptions described previously.

To reduce overall wing deflection, potential design modifications include the addition of a reinforcing strut to increase structural stiffness or altering the air foil characteristics to shift the lift distribution closer to the wing root.

Validation and Comparison

Due to the lack of resources on Spitfire wing deflection, a simplified case will be analytically solved using the principles of a cantilever beam, neglecting drag force. For this, the dual spar layout will be simplified to a single spar at the quarter chord with a representative moment of inertia as seen in Figure 14.

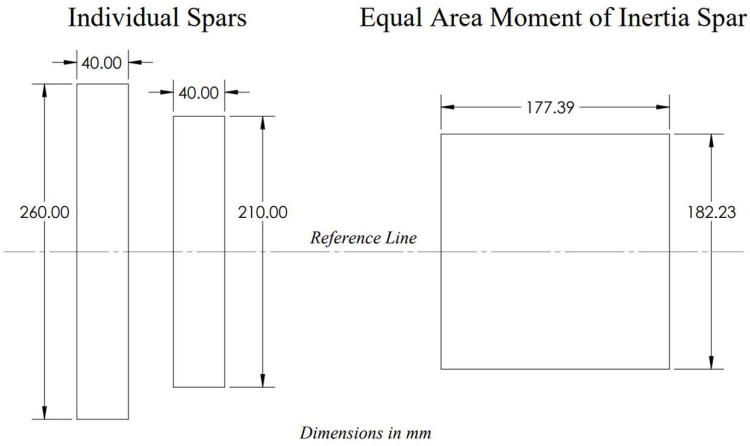


Figure 14: Equal area moment of inertia cross-section diagram

The area moment of inertia for a rectangle is given in *equation 2*, where b and h are the dimensions of the rectangle parallel and perpendicular with the reference plane respectively.

$$I = \frac{bh^3}{12} \qquad (3)$$

The equivalent spar was formulated using *equation 3*, keeping the height of the spar equal to the quarter-chord thickness of the root aerofoil. This was based on an area moment of inertia of 8.946 x 10^{-5} m^4. The next step is to calculate a representative loading. The same constant pressure can be applied along the bottom surface which is equal to the lift loading at the half-span. The calculations for this can be found in *appendix A2.*

The results of this analysis compared to the simulation results is summarised in table 5.

table 5: FEA results comparison

Parameter	Simulation Result	Hand Calculations
Max Stress	196 MPa	234.12 MPa
Max Deformation	210.51 mm	237.62 mm

The hand calculations marginally overestimate the stresses relative to the FEA results. This is likely because the analytic model omits discrete structural elements, such as ribs. that can redistribute loads and introduce torsional coupling between spars, reducing local bending stresses. Additionally, a convergence based solely on displacement can mask errors in peak stress prediction, particularly near stress concentrations or the previously encountered singularity.

To improve fidelity, a mesh-convergence study using stress metrics could be performed rather than displacement alone. Additionally, refining the mesh locally around the observed stress concentrations, and using more complex elements where appropriate, and removing or mitigating singularities could yield in more accurate results.

Discussion

For this specific wing, Figure 15 illustrates the calculated safety factor distribution. The structure remains within allowable limits under the applied loading, with a minimum safety factor of 1.6484. This result further confirms that the initial failure is expected to occur at the lower surface of each spar.

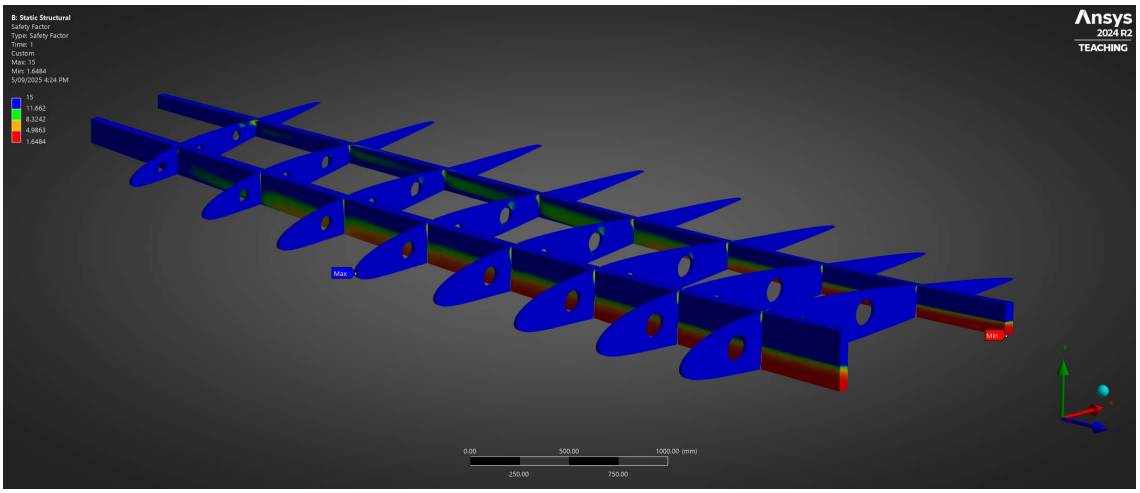


Figure 15: Factor of safety FEA results

Based on the results, the wing can be considered structurally acceptable with respect to the maximum stress observed. A factor of safety greater than 1.5 is typically regarded as sufficient for aviation applications, where minimizing structural weight remains a critical design driver [9]. The assessment of displacement is less straightforward. Unlike stress, there is no universal criterion for allowable wing deflection; instead, acceptable limits are defined on an aircraft-specific basis to ensure safe and predictable handling. Excessive deflection can alter aerodynamic characteristics and reduce control effectiveness, which is particularly critical during high-load manoeuvres such as the case examined in this study.

If fatigue life was required to be factored into this model, some changes would be required to accurately capture results relevant to a manufactured wing.

- The internal structure would have to be redesigned to better represent real manufacturing techniques such as including truss members, rivets and bolts
- The interactions between internal components will also have to be modelled with more complexity to better capture fatigue phenomena at joints and welds.
- The typical flight profile of the plane would have to be considered to accurately represent the cycling the wing would see.
- A skin would need to be included to accurately represent the torsional stiffness of the wing structure which would in turn affect the load cycling.

A keynote for the analysis conducted in this report is the lack of wing skin. For the Supermarine Spitfire this would have been an approximately 0.5 mm thin aluminium stressed skin plating that interfaces with the wing structure through both conventional and flush rivets

[2]. In additional to the complex truss structure of the wing, this skin would have supported the wing in torsional bending, creating boxed segments around the internal structure which offers increased rigidity [5]. The exclusion of this for the sake of simplicity may invalidate the torsional aspect of the total deformation results, as the maximum was seen at the trailing edge as a direct result of twisting. Despite this, the hand calculations yielding a similar stress validated the simulation for bending alone. It was previously noted the bending stresses may have been partially transferred to the torsional stresses in the structure, explaining the overshot hand calculations, yet regardless the structure did not fail under loading in both forms of analysis.

To further improve the FEA of this wing, more time would be required to accurately model the wing geometry, including skin, and more complex internal structures. A finer handling of the mesh within ANSYS would also be required to remove any erroneous results from problematic nodes.

Conclusion

This report presents the finite element analysis and validation of a wing structure inspired by the Supermarine Spitfire. It outlines the key assumptions, modelling approach, and meshing techniques used to set up the simulation within ANSYS. The analysis concluded that the representative wing geometry could safely withstand a 6g manoeuvre, with the most likely failure location identified on the underside of each spar, agreeing with the analytical predictions.

References

[1] ANSYS Inc, “ANSYS Homepage,” 2025. [Online]. Available: https://www.ansys.com/.

[2] Wikipedia, “Supermarine Spitfire,” 2025. [Online]. Available: https://en.wikipedia.org/wiki/Supermarine_Spitfire.

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[9] The Smithsonian, “Weight and Strength,” [Online]. Available: https://howthingsfly.si.edu/structures-materials/weight-and-strength-0.

A. Appendices

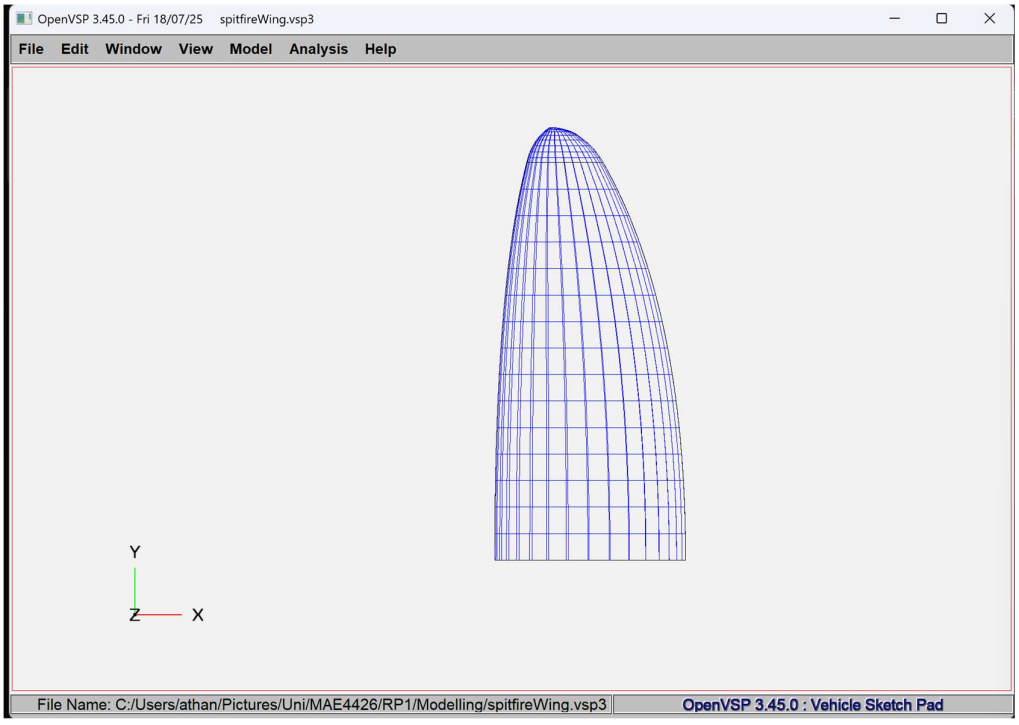
A1. Drag Determination

To obtain a drag coefficient we must find the value of induced and parasitic drag, fulfilling the equation

$$C_D = C_{d,0} + \frac{C_L^2}{\pi AR\epsilon}$$

The coefficient of lift can be obtained by simulating the wing geometry within OpenVSP.

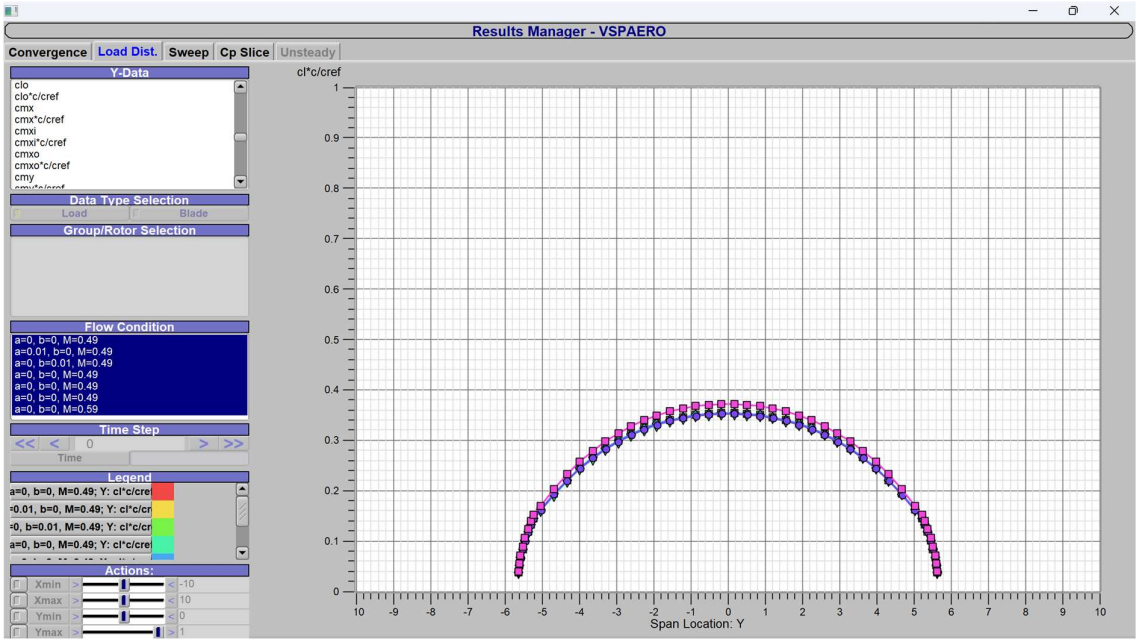
Wing model within OpenVSP



To find the Mach number required we can begin with the Spitfires top speed of 167 m/s. Assuming 11 km conditions (service ceiling), the speed of sound is 295.2 m/s. Hence the Mach number can be easily obtained via.

$$M = \frac{V}{a} = \frac{167}{295.2} = 0.566$$

This will be passed into the simulation in OpenVSP.



As expected, the lift distribution is elliptical. This simulation produced a result of 0.18 at 0 degrees angle of attack. To find a value of parasitic drag, the following empirical equation will be employed to find a coefficient of friction

$$C_f = \frac{0.455}{(\log_{10} R_e)^{2.58}(1 + 0.144M^2)^{0.65}}$$

With the previously obtained Mach number and a Reynolds number of 1e7 (to match OpenVSP simulation), a coefficient of friction of 0.002917 can be obtained. With an empirical correction factor of 1.27 for a lifting surface, and an assumption that the wetted surface is double the planform area (18.709 m^2). the final parasitic drag can be calculated

$$C_{d,0} = (1.27)(0.00217)(18.709) = 0.0693$$

With an aspect ratio of 5.4 and an efficiency factor of 1 (elliptical wings are awesome) the final drag coefficient can be calculated.

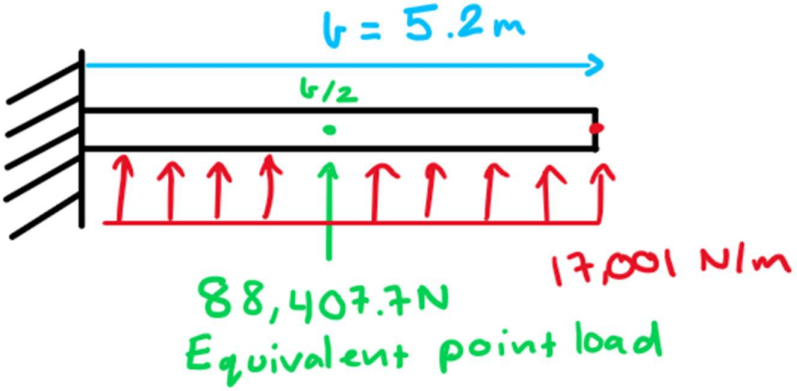
$$C_D = 0.0693 + \frac{0.18^2}{\pi(5.4)(1)} = 0.0712$$

This can then be used to calculate the drag force on the skin of the wing flying at 600 km/h at 11,000 metres where the air density is 0.3562 kg/m^3.

$$D = \frac{1}{2}\rho V^2 C_D S = \frac{1}{2}(0.3652)(167)^2(0.0712)(9.354) = \ 3392.31\ N$$

A2. Deformation and Stress Hand Calculations

UNIFORM LOADING



MOMENT AT ROOT

$$M = F_x = (88{,}407.7\,N)(2.6)$$
$$= 229{,}860\ N/m$$

TIP DEFLECTION

$$\delta_{tip} = \frac{w b^4}{8EI} = \frac{(17{,}001)(5.2)^4}{8(73.1\times10^9)(8.946\times10^{-5})}$$

$$\boxed{\delta_{tip} = 237.618\ mm}$$

MAX STRESS AT ROOT

$$\sigma = \frac{My}{I}\ ;\ at\ y = 0.091\ m$$

$$\sigma = \frac{(229{,}860)(0.091)}{8.946\times10^{-5}}$$

$$\boxed{\sigma_{max} = 234.12\ MPa}$$